TERM DEPOSIT CAMPAIGN

Six contacts, one conversion target

100

Predict

Score held-out customers

10

Optimize

Choose six under two limits

6

Simulate

Estimate at least three sales

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STAGE 1

Predict subscription probabilities

Hybrid classifier

Train-only scaling and encoding

5-qubit RY and RZZ feature map

48 fidelity features + logistic regression

300 labeled rows, stratified 5-fold validation

Validation	Classical	Hybrid
ROC AUC	0.5856	0.5933
Average precision	0.2016	0.2079
Brier score	0.1020	0.1018
Precision@10	0.40	0.30

Rank 100 test probabilities. The top ten advance, with ties broken by customer ID.

STAGE 2

Choose six contacts under both limits

Maximize expected deposit value minus contact cost

Classical	Build the cost function and check all 210 six-customer subsets.
Quantum	Run two QAOA layers on ten qubits and measure 32,768 shots.
Decode	Keep the highest-scoring feasible measured string, 1100011101.

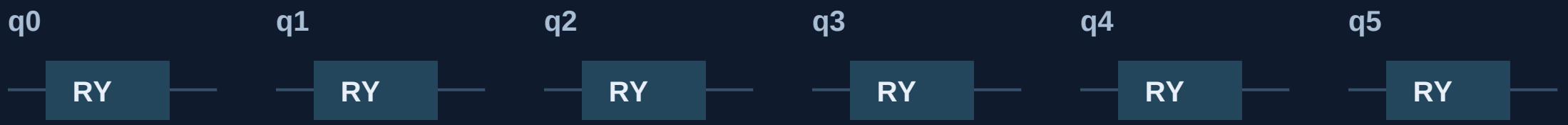
C0031 C0079 C0005 C0318 C0327 C0170

Cost 12/25 Capacity 15/22 Net value 23.01459 Gap to classical optimum 0

STAGE 3

Estimate the chance of three sales

One RY rotation per customer



32,768 shots

22.65%

Exact 22.92% / Monte Carlo 22.85%

3% readout flips: 24.93% raw, 22.51% corrected

Model assumes independent responses. Treat the estimate as a pilot-planning input.